CHILD NUTRITION ASSESSMENT REPORT

Agentic AI Framework for Explainable Child Nutrition Recommendation and Malnutrition Prevention

1. Child Information

Measurement	Value
Age	22 months
Weight	10.00 kg
Height	82.00 cm
MUAC	13.50 cm
BMI	14.87

2. Nutrition Risk Prediction

Parameter	Result
Predicted nutritional status	normal
Best machine learning model	LightGBM

3. Prediction Probabilities

Nutritional Status	Probability
moderate	0.00%
normal	100.00%
severe	0.00%

4. Personalized Nutrient Priorities

Nutrient	Priority Weight
Protein	1.00
Calcium	0.90
Iron	0.90
Dietary Fiber	0.75
Vitamin A	0.75
Vitamin C	0.70

5. Recommended Child-Suitable Foods

Recommended Food	Priority Score	Nutrient Information
rice bran	—	—
soybean curd cheese	—	—

Recommended Food	Priority Score	Nutrient Information
breadnut tree seeds dried	—	—
mustard spinach cooked	—	—
chickpeas cooked	—	—
yogurt low fat	—	—
orange roughy cooked	—	—
peanuts cooked	—	—
navy beans cooked	—	—
lentils cooked	—	—

6. Nutrients in Recommended Foods

Food	Protein	Iron	Calcium	Zinc	Vitamin A	Vitamin C
rice bran	15.80	21.90	67.30	7.10	0.00	0.00
soybean curd cheese	28.10	12.60	423.00	3.90	0.00	0.00
breadnut tree seeds dried	13.80	7.40	150.40	3.10	0.00	74.60
mustard spinach cooked	3.10	1.40	284.40	0.20	0.00	117.00
chickpeas cooked	14.50	4.70	80.40	2.50	0.00	2.10
yogurt low fat	12.90	0.20	448.40	2.20	0.02	2.00
orange roughy cooked	6.40	0.30	3.10	0.10	0.05	0.00
peanuts cooked	24.30	1.80	99.00	3.30	0.00	0.00
navy beans cooked	15.00	4.30	125.60	1.90	0.00	0.00
lentils cooked	17.90	0.50	0.08	0.01	0.00	0.40

7. Indian Food Images and Image Matching

Recommended Food	Image Food	Image Relation
rice bran	rice_bran	exact
soybean curd cheese	soyabean_curd_chease	fuzzy
breadnut tree seeds dried	breadnut_tree_seeds_dried	exact
mustard spinach cooked	mustard_spinach_cooked	exact
chickpeas cooked	chickpeas_cooked	exact
yogurt low fat	yougurt_low_fat	fuzzy
orange roughy cooked	orange_roughy_cooked	exact
peanuts cooked	peanuts_cooked	exact
navy beans cooked	navy_beans_cooked	exact
lentils cooked	lentils_cooked	exact

Selected Indian Food Images

Recommended food: rice bran

Image food: rice_bran

Image relation: exact



Recommended food: soybean curd cheese

Image food: soyabean_curd_chease

Image relation: fuzzy



Recommended food: breadnut tree seeds dried

Image food: breadnut_tree_seeds_dried

Image relation: exact



Recommended food: mustard spinach cooked

Image food: mustard_spinach_cooked

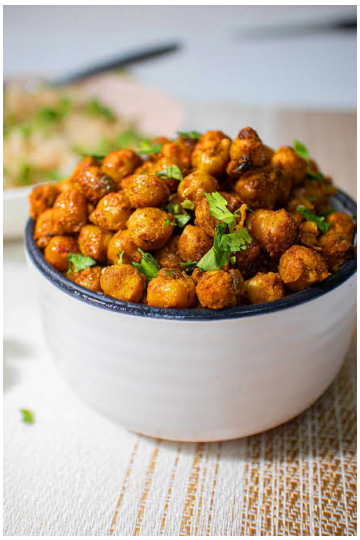
Image relation: exact



Recommended food: chickpeas cooked

Image food: chickpeas_cooked

Image relation: exact



Recommended food: yogurt low fat

Image food: yougurt_low_fat

Image relation: fuzzy



Recommended food: orange roughy cooked

Image food: orange_roughy_cooked

Image relation: exact



Recommended food: peanuts cooked

Image food: peanuts_cooked

Image relation: exact



Recommended food: navy beans cooked

Image food: navy_beans_cooked

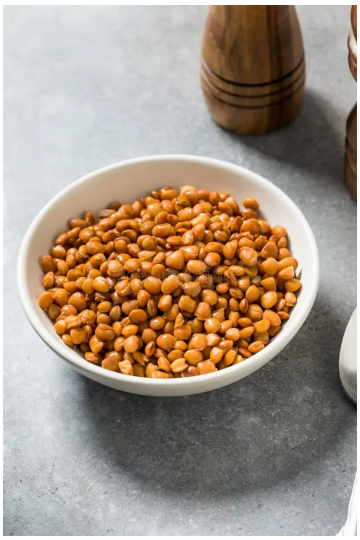
Image relation: exact



Recommended food: lentils cooked

Image food: lentils_cooked

Image relation: exact



8. Personalized Meal Plan

Meal	Recommended Food	Priority Nutrients	Nutrient Information
Breakfast	rice bran	protein, calcium, iron, fiber, vitamin_a, vitamin_c	energy: 373.00; protein: 15.80; fat: 24.60; carbohydrate: 58.60; fiber: 24.80; iron: 21.90; calcium: 67.30; zinc: 7.10; vitamin_a: 0.00; vitamin_b12: 0.00; vitamin_c: 0.00; vitamin_d: 0.00
Mid-Morning	soybean curd cheese	protein, calcium, iron, fiber, vitamin_a, vitamin_c	energy: 340.00; protein: 28.10; fat: 18.20; carbohydrate: 15.50; fiber: 0.00; iron: 12.60; calcium: 423.00; zinc: 3.90; vitamin_a: 0.00; vitamin_b12: 0.00; vitamin_c: 0.00; vitamin_d: 0.00
Lunch	rice bran	protein, calcium, iron, fiber, vitamin_a, vitamin_c	energy: 373.00; protein: 15.80; fat: 24.60; carbohydrate: 58.60; fiber: 24.80; iron: 21.90; calcium: 67.30; zinc: 7.10; vitamin_a: 0.00; vitamin_b12: 0.00; vitamin_c: 0.00; vitamin_d: 0.00
Evening Snack	soybean curd cheese	protein, calcium, iron, fiber, vitamin_a, vitamin_c	energy: 340.00; protein: 28.10; fat: 18.20; carbohydrate: 15.50; fiber: 0.00; iron: 12.60; calcium: 423.00; zinc: 3.90; vitamin_a: 0.00; vitamin_b12: 0.00; vitamin_c: 0.00; vitamin_d: 0.00
Dinner	rice bran	protein, calcium, iron, fiber, vitamin_a, vitamin_c	energy: 373.00; protein: 15.80; fat: 24.60; carbohydrate: 58.60; fiber: 24.80; iron: 21.90; calcium: 67.30; zinc: 7.10; vitamin_a: 0.00; vitamin_b12: 0.00; vitamin_c: 0.00; vitamin_d: 0.00

9. Explainable AI – SHAP Prediction Explanation

Explained prediction: normal

Feature	SHAP Contribution
SHAP feature contribution data	See generated SHAP explanation

10. Personalized Nutrition Advice and Malnutrition Prevention

Agent

Nutrition Advice / Malnutrition Prevention Agent

Priority Profile

normal

Age Based Advice

• Provide a varied diet containing energy- and nutrient-dense foods.

• Include suitable protein-rich foods regularly.

• Include iron-rich foods together with vitamin-C-rich foods.

• Offer a variety of fruits and vegetables.

• Provide age-appropriate meals and snacks throughout the day.

Nutrient Advice

Food Based Advice

• Prioritized foods include: rice bran, soybean curd cheese, breadnut tree seeds dried, mustard spinach cooked, chickpeas cooked.

Meal Plan Advice

• Include a nutrient-dense breakfast appropriate for the child's age.

• Include a balanced lunch containing suitable energy, protein and micronutrient sources.

• Provide a suitable nutrient-dense evening meal.

Prevention Advice

• Maintain a varied and balanced diet.

• Continue regular age-appropriate meals and snacks.

• Include a variety of fruits, vegetables, grains and protein sources.

• Continue monitoring growth and nutrition status.

All Advice

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11. Assessment Summary

The assessment considered the child's age, weight, height and MUAC measurements. The calculated BMI was 14.87. The machine-learning prediction was normal, using LightGBM as the selected model. The agentic pipeline then generated personalized nutrient priorities, child-suitable food recommendations, food-image matching, image validation, a meal plan, SHAP-based explanation and nutrition guidance.